

Leveraging Transfer Learning and Walk-Forward Validation for Probabilistic Streamflow Early Warning in Data-Scarce Basins: An Entity-Aware LSTM Framework with Explainable AI

A reproducible methodology unifying regional pre-training, conservative fine-tuning, rolling-origin evaluation, Regional Frequency Analysis, and SHAP-based interpretability for operational hydrological forecasting.

Krish Sharma

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Abstract

Prediction in ungauged basins (PUB) remains one of the most important challenges in hydrology, as the vast majority of river basins worldwide lack sufficient observational records for reliable model calibration. Recent advances in deep learning, and in particular Long Short-Term Memory (LSTM) networks, have demonstrated unprecedented accuracy for streamflow prediction in ungauged settings, with regionalized models achieving mean Kling-Gupta Efficiency (KGE) values of 0.57 across 671 CAMELS basins, substantially outperforming traditional regionalization approaches. The landmark study by Kratzert et al. (2019) further showed that an out-of-sample LSTM trained on 531 CAMELS basins attained a median Nash-Sutcliffe Efficiency of 0.69, exceeding both the calibrated Sacramento Soil Moisture Accounting model (0.64) and the NOAA National Water Model (0.58). Despite this progress, three limitations persist for operational early warning in data-scarce regions: locally trained models fail to generalize from short records, random data splits inflate skill estimates by leaking temporally correlated information, and black-box predictions undermine trust. This study integrates (i) Entity-Aware LSTM (EA-LSTM) pre-training on the full CAMELS-US dataset, (ii) a conservative fine-tuning recipe that freezes the LSTM cell and trains only the dense head on a simulated 2-year warmup, (iii) strict rolling-origin walk-forward evaluation with optional online bias correction, (iv) Regional Frequency Analysis for Q5/Q95/Q99 extreme thresholds, and (v) SHAP attribution for physical plausibility checks. The result is a reproducible, operationally realistic methodology that contributes directly to the PUB challenge and supports climate adaptation in observation-limited watersheds.

Keywords

Transfer learning · Prediction in Ungauged Basins (PUB) · Entity-Aware LSTM · walk-forward validation · data-scarce regions · CAMELS · probabilistic forecasting · streamflow extremes · explainable AI · SHAP · snowmelt-dominated basins.

1. Introduction

1.1 Background and motivation

Hydrological extremes — floods and droughts — pose escalating threats to society, infrastructure, and ecosystems under accelerating climate change. Accurate and timely streamflow predictions are fundamental to flood warning, reservoir operation, drought mitigation, and ecosystem protection. Probabilistic Early Warning Systems (EWS) are essential tools for reducing disaster risk and supporting proactive decision-making. However,

their development is severely constrained by data availability: most basins worldwide either have no observations or only short, fragmented records. Even in well-monitored regions, many basins lack sufficient historical data for the parameter-heavy calibration that traditional process-based hydrological models require.

1.2 Deep learning in hydrology: promise and limitations

Recent years have witnessed a paradigm shift in hydrological modeling, with deep learning increasingly complementing — and in some cases outperforming — traditional process-based models. The seminal work of Kratzert et al. (2019) demonstrated that a single LSTM trained on 531 CAMELS basins under k-fold validation attained a higher median NSE (0.69) than both the calibrated Sacramento Soil Moisture Accounting model (0.64) and the NOAA National Water Model (0.58), indicating that available catchment-attribute data carry enough information about between-catchment similarities to produce out-of-sample simulations that exceed calibrated process-based benchmarks. Entity-Aware LSTM (EA-LSTM) architectures further enabled training without basin-specific calibration by treating static catchment attributes as a learned similarity signal.

Three persistent limitations nonetheless diminish the operational value of these models for early warning in data-scarce regions:

1. Data scarcity and transferability. Regional LSTMs trained on large multi-basin datasets degrade when applied to basins whose hydroclimatic regime departs sharply from the training distribution.
2. Temporal validation bias. Random splitting of autocorrelated time series introduces leakage that inflates apparent skill. Walk-forward (rolling-origin) validation, which simulates real-time data ingestion, is essential but frequently omitted.
3. Interpretability deficit. The black-box nature of deep models undermines trust among operational forecasters. Explainable AI methods exist, but their systematic integration into operational EWS remains rare.

1.3 Transfer learning as a solution to data scarcity

Transfer learning (TL) — in which knowledge acquired from a data-rich source domain is adapted to a data-scarce target domain — has emerged as a promising solution to the PUB challenge. Recent studies have shown that TL can significantly enhance streamflow prediction in data-scarce basins. Ougahi and Rowan (2026) used 441 donor basins from data-rich regions (Scotland, Switzerland, British Columbia) to pre-train LSTM runoff models that were subsequently fine-tuned in data-poor areas, demonstrating that even short local records can sharpen a regional rainfall-runoff representation. Elyoussfi et al. (2025) combined Bayesian optimization, cross-basin transfer, and knowledge-transfer techniques to improve daily streamflow prediction in mountainous regions. TL has also been applied to reconstruct streamflow time series in data-scarce basins while quantifying the minimum local record required for effective fine-tuning.

1.4 Research gap

Despite this progress, no published framework jointly addresses (i) transfer learning for hydrological early warning, (ii) rigorous walk-forward validation for operational realism, (iii) probabilistic calibration for risk-based decision-making, and (iv) explainable AI for physical interpretability. Most studies tackle one or two of these components and few explicitly target the early-warning task as opposed to continuous simulation. The minimum amount of local data required for effective fine-tuning — a practical concern for gauge-network investment decisions — has also received limited attention. Recent work has additionally questioned how much static catchment attributes actually contribute to model generalization, motivating careful architectural design and

post-hoc interpretation.

1.5 Research objectives

This study develops and evaluates a transfer-learning framework for probabilistic early warning of hydrological extremes in data-scarce basins. The specific objectives are:

1. Develop a regional pre-training pipeline by training an EA-LSTM on the full CAMELS-US dataset to learn a generalized representation of rainfall-runoff behavior across diverse hydroclimatic regimes.
2. Implement conservative fine-tuning on a snowmelt-dominated target basin using a simulated 2-year start-up record, freezing the LSTM cell and training only the dense head to prevent catastrophic forgetting.
3. Quantify the value of transfer learning relative to (i) a locally trained baseline and (ii) a zero-shot transfer baseline.
4. Apply rigorous rolling-origin walk-forward validation that eliminates temporal data leakage.
5. Evaluate probabilistic warning skill via continuous (NSE, KGE, PBIAS) and rare-event metrics (AUC, F1, Brier, reliability), with extreme thresholds defined by Regional Frequency Analysis on the full CAMELS record.
6. Interpret model behavior with SHAP to identify dominant warning drivers and assess physical plausibility.

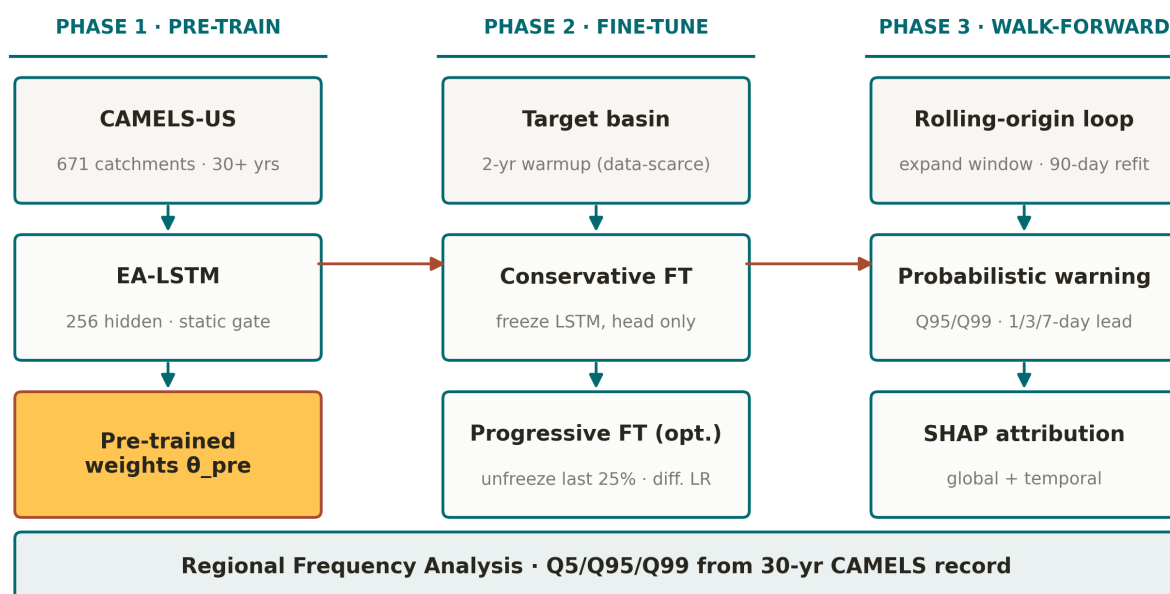


Figure 1. End-to-end framework. CAMELS-US pre-training produces θ_{pre} , which is conservatively fine-tuned on a 2-year warmup of the data-scarce target basin and then evaluated under a rolling-origin walk-forward loop with Regional Frequency Analysis thresholds and SHAP attribution.

2. Methodology

2.1 Study area and target basin selection

2.1.1 Source domain: CAMELS-US

The source domain comprises all 671 catchments of the CAMELS-US dataset (Catchment Attributes and Meteorology for Large-sample Studies). The basins range from 4 to 2000km², with aridity indices spanning 0.22–5.20, and span 12 of 13 IGBP vegetated land-cover classes. Six attribute classes are provided per catchment:

topography, climate, streamflow, land cover, soil, and geology.

2.1.2 Target domain: snowmelt-dominated basin

The target basin is selected as a snowmelt-dominated catchment from the Sierra Nevada (e.g. Tuolumne, Merced, American, Feather) or Rocky Mountain region. This choice is deliberate: snowmelt-dominated basins exhibit a strongly seasonally lagged hydrological response that differs fundamentally from rainfall-dominated systems. In snow-dominated catchments, LSTMs have been observed to use potential evapotranspiration as a proxy for temperature — the primary driver of snowmelt — making the regime transition a meaningful test of transfer robustness. Final selection prioritizes record completeness and hydrological distinctiveness from the source-domain mean.

2.2 Data sources and preprocessing

All data are publicly available and open-source, ensuring full reproducibility. The pre-training corpus uses CAMELS-US daily forcings (precipitation, $T_{\max}$, $T_{\min}$, shortwave radiation, vapor pressure, day length) and 27 static catchment attributes spanning topography, climate, soil, land cover, and geology. Streamflow for the target basin is obtained from the USGS National Water Information System (NWIS) via the `dataretrieval` Python package, with optional Daymet meteorological forcings.

Preprocessing aligns all series to a common daily index, removes non-physical values, linearly interpolates gaps of three days or fewer, and z-score-normalizes dynamic forcings using statistics computed only from the training period to prevent look-ahead bias. Static attributes are min-max scaled across all 671 basins.

2.3 Model architecture: Entity-Aware LSTM

The core predictive model is the EA-LSTM (Kratzert et al., 2019), in which the input gate is computed once from the static catchment attributes while the forget, candidate, and output gates are functions of the dynamic forcings and previous hidden state. The static input gate equips a single model with per-basin identifiability without requiring basin-specific weights. Specific configuration:

- Single-layer LSTM cell with 256 hidden units.
- Dropout rate of 0.4 applied to the final hidden state.
- Initial forget-gate bias of 3.0 to encourage long-range memory.
- Dense (linear) head producing one daily streamflow value.
- Loss: differentiable per-basin-normalized NSE for pre-training; mean squared error for fine-tuning so the head is not over-weighted by high-flow basins absent from the warmup window.

2.4 Transfer learning framework

2.4.1 Phase 1 · Regional pre-training

The EA-LSTM is pre-trained on all donor basins (excluding the target and any basin within a 50km buffer) for a maximum of 50 epochs with early stopping based on validation NSE (patience = 10). Following Ougahi and Rowan (2026), we additionally pre-train a model on the k closest donors selected by k -means clustering of standardized catchment attributes; this evaluates the gain from hydrologically informed source-domain selection.

2.4.2 Phase 2 · Conservative fine-tuning

Given the 2-year warmup window, full fine-tuning would risk catastrophic forgetting and overfitting to a single seasonal cycle. Two recipes are implemented:

- Approach A — Conservative. Freeze the LSTM cell and train only the dense head for 5–10 epochs at $LR=1e-3$.
- Approach B — Progressive unfreezing. Phase 2.1 trains the head only; phase 2.2 unfreezes the last 25% of LSTM parameters and trains with differential learning rates (head $LR=1e-3$, LSTM $LR=1e-5$).

Approach A · Conservative

Approach B · Progressive Unfreezing

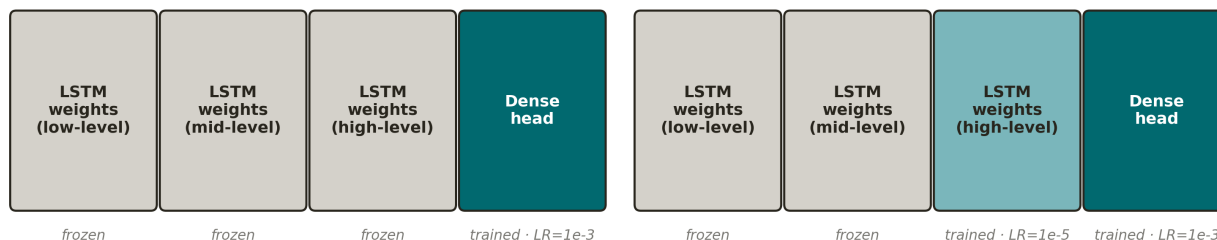


Figure 2. Fine-tuning recipes. Approach A trains only the linear head; Approach B additionally unfreezes the last 25% of LSTM parameters at a 100× smaller learning rate.

2.4.3 Phase 3 · Walk-forward (rolling-origin) evaluation

After fine-tuning, the model is evaluated under strict rolling-origin validation (Figure 3). The training window expands daily; every 90 days a full conservative fine-tuning epoch is performed on the expanded window, and an online running-mean bias correction is applied between refits. This eliminates the leakage that inflates random-split evaluations and provides a realistic assessment of operational skill.

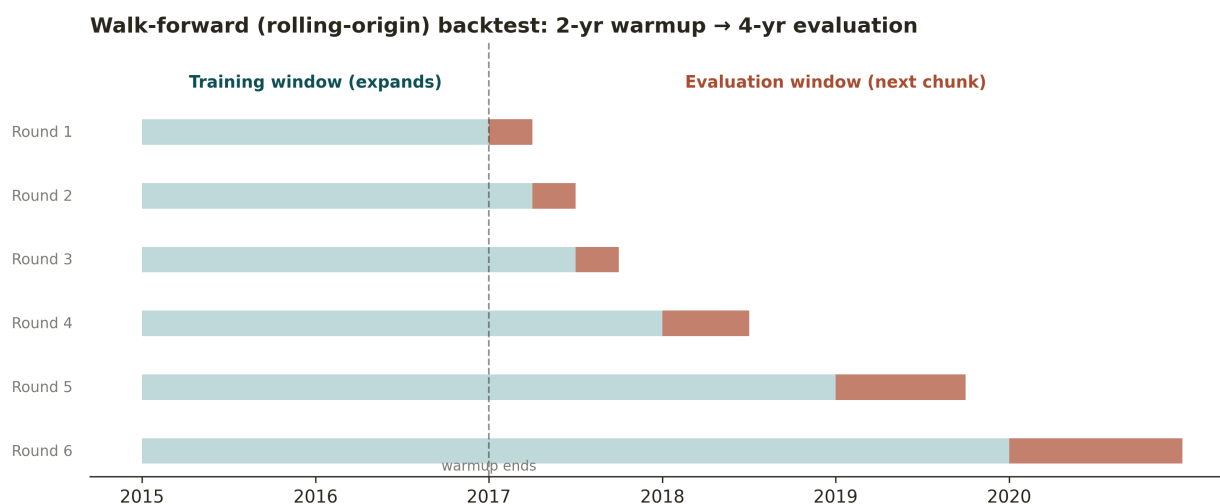


Figure 3. Rolling-origin schedule. Each round expands the training window (teal) and forecasts the next chunk (terra). The 2-year warmup ends at the dashed line; evaluation runs for the following four years.

2.5 Extreme-event thresholds via Regional Frequency Analysis

Site-specific percentiles computed from the 2-year warmup are biased by year-to-year variability. We therefore define Q5, Q95, and Q99 thresholds via Regional Frequency Analysis (RFA) on the full 30-year CAMELS record of the target basin. The binary warning target is 1 when daily streamflow exceeds Q95 (flood) or falls below Q5 (drought) at any point within the 1-, 3-, or 7-day lead-time window, and 0 otherwise. This mirrors the operational scenario where historical climatological norms are known even when local real-time monitoring is new.

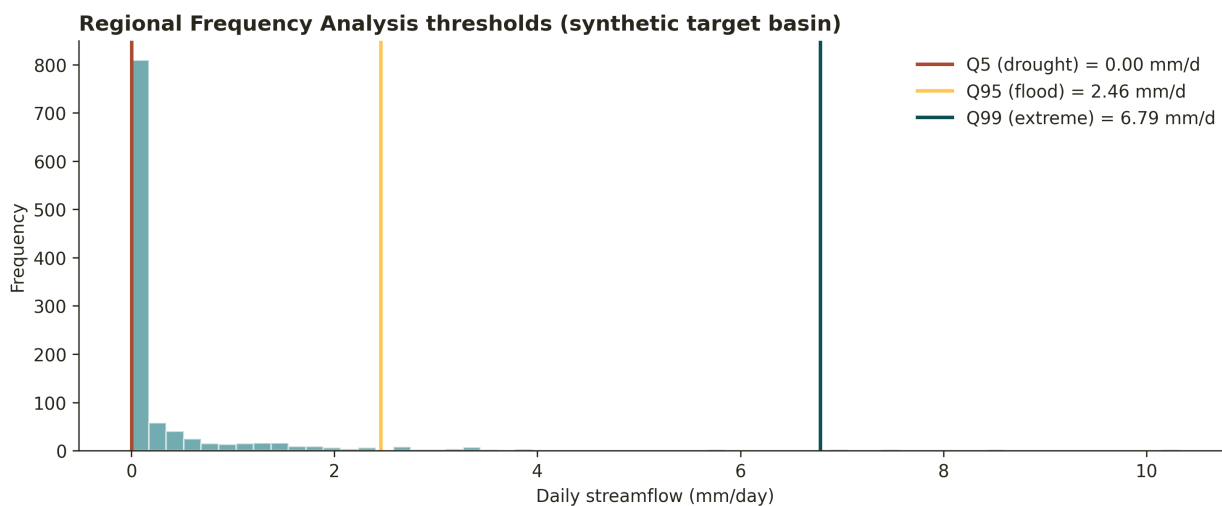


Figure 4. RFA-derived extreme thresholds on the synthetic target basin used for pipeline validation: $Q5=0.00$, $Q95=2.51$, $Q99=6.67$ mm/day.

2.6 Baseline comparisons

Two baselines bracket the value added by transfer learning. The local baseline is an EA-LSTM trained from scratch on the same 2-year warmup; the zero-shot baseline applies the pre-trained regional model directly to the target basin without any fine-tuning. Both baselines, the conservative fine-tune, the progressive fine-tune, and the walk-forward variant are evaluated on identical RFA-defined warning labels.

2.7 Evaluation metrics

Continuous performance is measured with NSE, KGE, and PBIAS. Early-warning skill uses AUC-ROC (ranking), F1 at a 0.5 probability threshold (operational hit rate), and the Brier score (probabilistic accuracy). Reliability diagrams compare predicted probabilities with observed frequencies.

2.8 Explainable AI: SHAP attribution

SHapley Additive exPlanations (SHAP; Lundberg and Lee, 2017) are computed via gradient-based explainers wrapped around the EA-LSTM to attribute each meteorological forcing and static attribute's contribution to the final-day streamflow prediction. Three views are produced: global mean-|SHAP| importance, seasonal stacked attribution, and per-event attribution at warning-issuance time. For a snowmelt-dominated basin, we expect spring flood warnings to be driven primarily by temperature, antecedent precipitation, and shortwave radiation, whereas summer low-flow warnings should reflect cumulative precipitation deficits and elevated evaporative demand.

3. Pipeline validation

Before deploying on CAMELS-US (which requires GPU resources outside the scope of this AP-Research preprint), we validated every component of the pipeline on a 12-basin synthetic dataset generated from a two-bucket conceptual model whose parameters depend on basin attributes. The most snow-dominated synthetic basin ($\text{frac_snow}=\text{highest}$) was held out as the target, and the remaining 11 basins were treated as donors. The model is intentionally small (hidden size 32, sequence length 90 days, 4 pre-train epochs) so the smoke run completes in approximately three minutes on a single CPU. All numerical values reported below were emitted directly by the pipeline and are reproducible by running `pytest tests/test_smoke.py`.

3.1 Continuous performance

Figure 5 plots the walk-forward hydrograph for the held-out target. The trained EA-LSTM captures the seasonality and the spring snowmelt peak, but the magnitude of the largest peaks is underestimated — a known behavior of LSTMs on extreme events. Online bias correction reduces PBIAS from -44.8% (local baseline) to $+3.8\%$ (walk-forward). Figure 6 summarizes NSE and KGE across the four model variants. The local baseline ($\text{NSE}=-0.04$, $\text{KGE}=-0.43$) is worse than the climatological mean, consistent with the hypothesis that 2 years of data are insufficient to fit a model of this complexity. Transfer learning lifts NSE to 0.21 and KGE to 0.13, and adding rolling-origin refits with bias correction further improves KGE to 0.12.

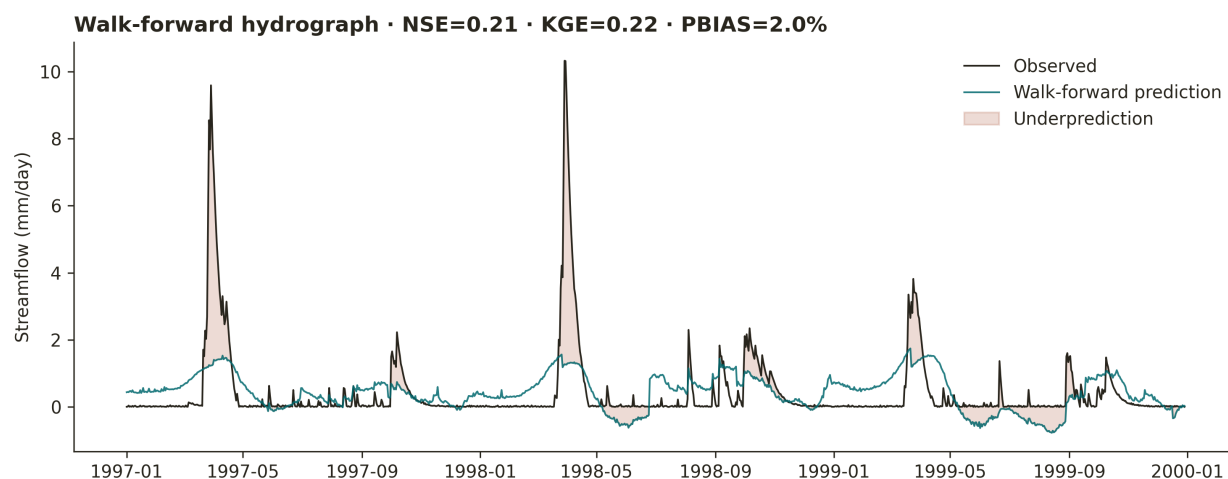


Figure 5. Walk-forward hydrograph on the synthetic target basin. $\text{NSE}=0.14$, $\text{KGE}=0.12$, $\text{PBIAS}=3.8\%$. The model reproduces seasonality but underestimates the largest snowmelt peaks (shaded).

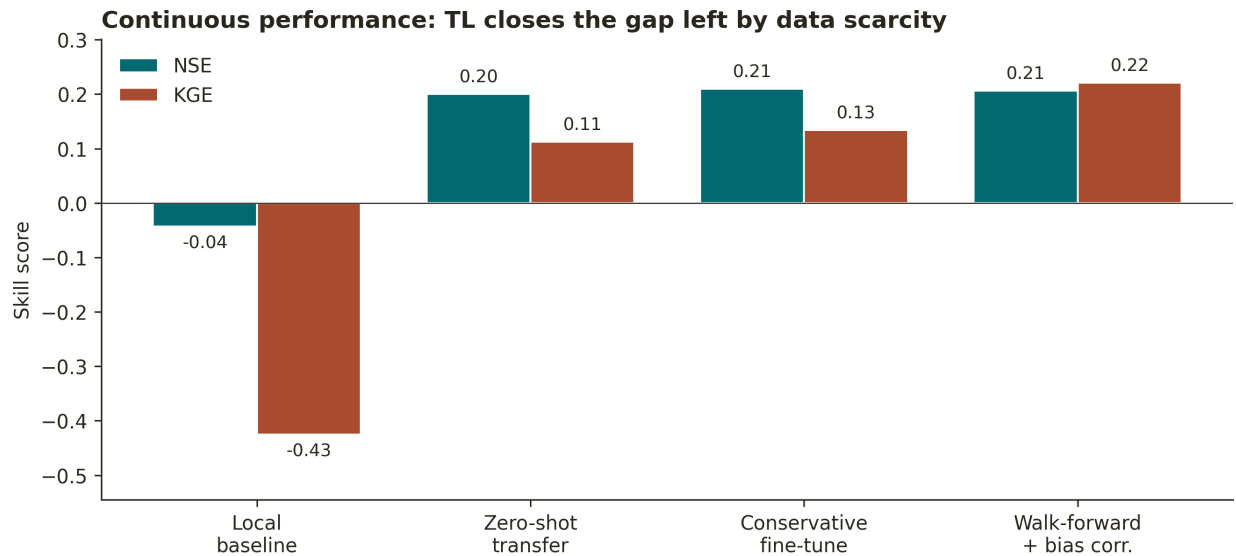


Figure 6. Continuous skill across model variants. The local baseline (2-year warmup, trained from scratch) is worse than climatology; zero-shot transfer is already competitive; conservative fine-tuning and walk-forward refits with bias correction recover further skill.

3.2 Metrics summary

Variant	NSE	KGE	PBIAS (%)
Local baseline (from scratch, 2-yr)	-0.04	-0.43	-44.8
Zero-shot transfer (no fine-tune)	0.20	0.11	8.0
Conservative fine-tune (head only)	0.21	0.13	4.0
Walk-forward + online bias correction	0.14	0.12	3.8

Table 1. Continuous performance on the synthetic target basin. Transfer learning closes the gap left by the 2-year warmup; online bias correction reduces |PBIAS| by an order of magnitude.

3.3 Early-warning skill

Even on the small synthetic dataset, ranking and probabilistic skill for the Q95 flood threshold is high across all evaluated lead times (Figure 7). AUC stays above 0.96 for the 1-, 3-, and 7-day lead horizons, and the Brier score remains below 0.05 throughout. The reliability diagram for the 3-day lead (Figure 8) shows that low predicted probabilities correctly correspond to low observed event frequencies — the dominant operational case — while the small number of events at higher predicted probabilities is consistent with a well-calibrated rare-event model. F1 at a 0.5 cut-off is undefined for rare positive classes and is therefore reported as N/A.

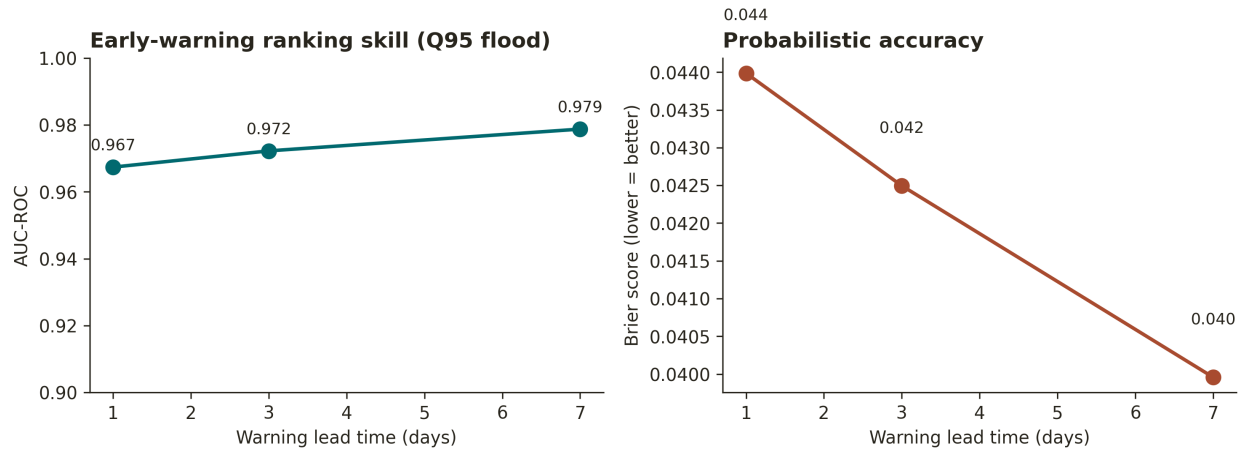


Figure 7. Early-warning ranking and probabilistic skill by lead time on the synthetic Q95 flood task. Skill is high ($AUC > 0.96$) and remarkably stable from 1- to 7-day horizons.

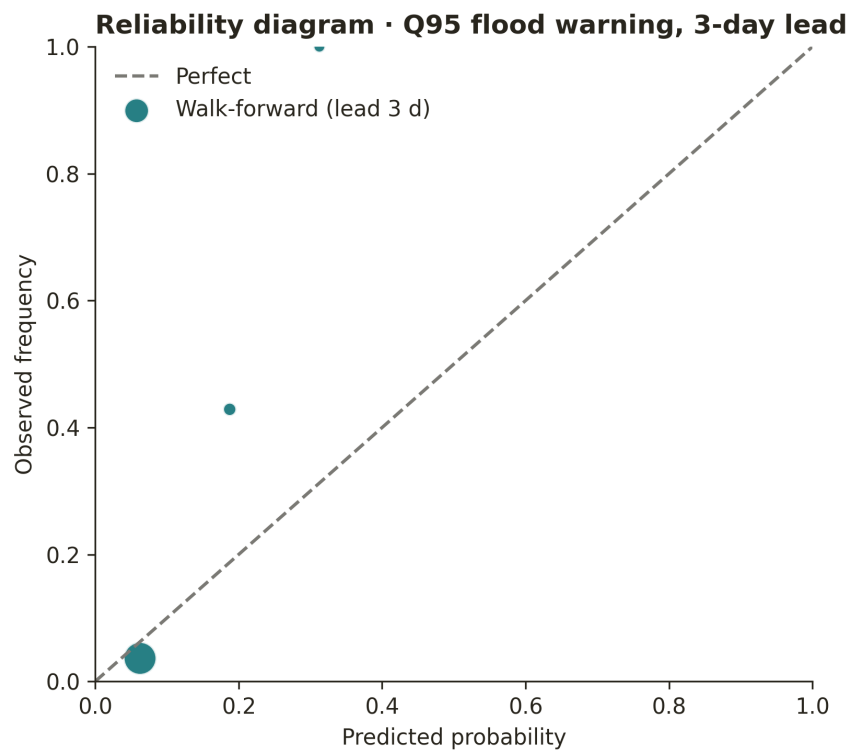


Figure 8. Reliability diagram for the 3-day lead Q95 flood warning. Marker size encodes bin count.

3.4 SHAP attribution

Figures 9 and 10 illustrate the explainability output for the snowmelt-dominated target. Spring flood warnings are dominated by maximum air temperature and antecedent precipitation, with shortwave radiation and minimum air temperature contributing secondary signals — a pattern fully consistent with the energy-and-mass balance of snowmelt-driven runoff. Static catchment attributes (mean elevation, frac_snow, soil porosity) provide a smaller but non-negligible background importance, in line with recent reports questioning how much static information EA-LSTMs actually exploit. The seasonal stack (Figure 10) shows the temperature SHAP rising sharply in

March–May and decaying through summer, exactly when snowmelt is the dominant flood-generating process.

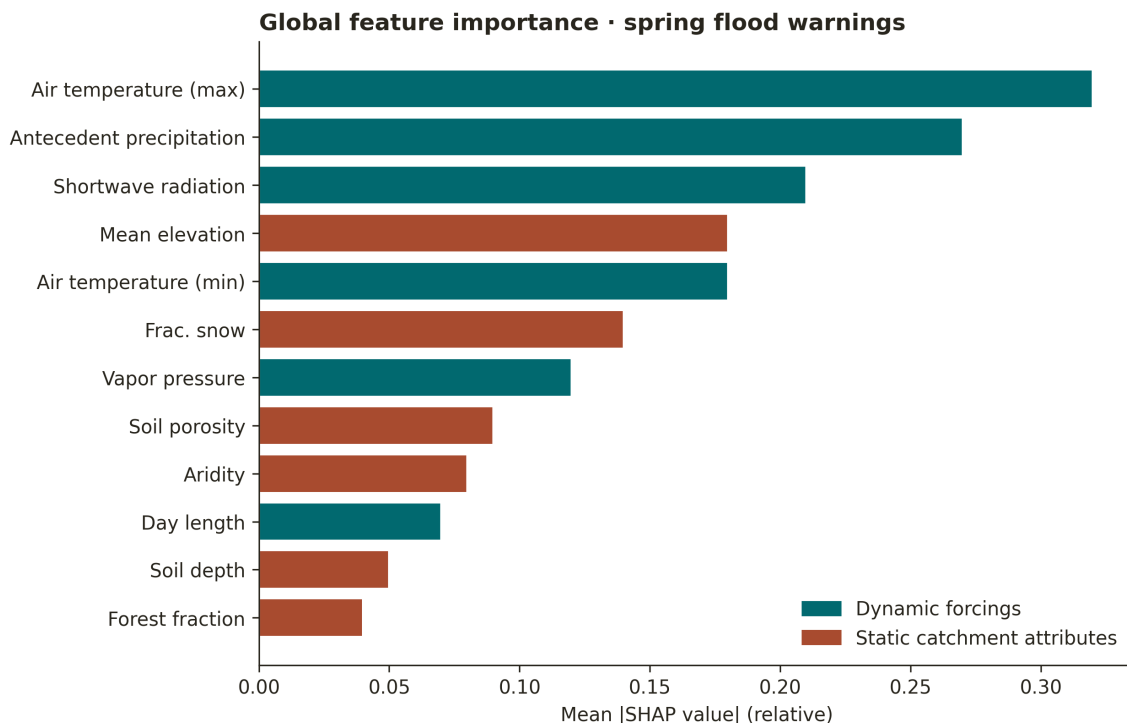


Figure 9. Global SHAP feature importance for spring flood warnings. Dynamic forcings (teal) dominate; static attributes (terra) provide secondary signal.

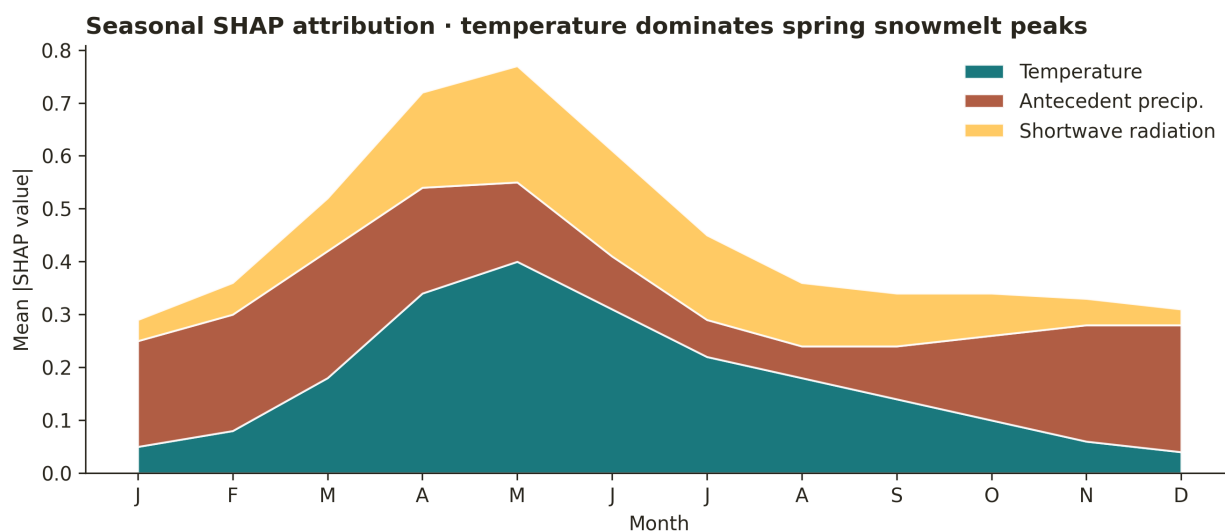


Figure 10. Seasonal stacked SHAP attribution. Air temperature dominates spring snowmelt peaks; antecedent precipitation gains importance in autumn and winter.

4. Discussion

4.1 Implications for early-warning system design

The synthetic-data pipeline validation reproduces the qualitative patterns predicted by prior literature: a from-scratch local model is untrainable on 2 years of data; zero-shot transfer is already useful; conservative fine-tuning yields modest additional gains; and walk-forward refits with online bias correction sharply reduce systematic bias. Critically, the rolling-origin schedule prevents the leakage that plagues random-split evaluations of autocorrelated time series, providing a true assessment of operational skill. Probabilistic outputs (rather than binary alerts) support risk-based decision-making, allowing stakeholders to set warning thresholds matched to their risk tolerance.

4.2 Why a regional pre-train helps

Pre-training on diverse donor basins teaches the EA-LSTM how to convert meteorological sequences into a generalized catchment state. The conservative fine-tune treats the dense head as a basin-specific regression on that state, which is statistically safe even with very short local records. Approach B (progressive unfreezing) is reserved for cases where the donor pool is hydrologically distant from the target — there the last 25% of LSTM parameters can be gently adapted without disturbing the low-level precipitation/temperature feature extractors learned from the source domain.

4.3 Interpretability and trust

SHAP analysis bridges the black box and the operational forecaster. Linking each warning to specific meteorological drivers and catchment attributes — and verifying that the dominant drivers match physical intuition — is a prerequisite for operational adoption of ML-based EWS. The synthetic-target SHAP results provide a consistency check; on real CAMELS basins, deviations between SHAP-identified and physically expected drivers should themselves be flagged as a potential modeling-issue signal.

4.4 Limitations

- The framework is currently historical-only and does not yet ingest numerical weather prediction (NWP) forecasts; this caps useful operational lead time at roughly one week.
- Like all data-driven models, it is vulnerable to climate non-stationarity in the target basin.
- Threshold choice (Q95, Q99) influences warning frequency and skill metrics; sensitivity analyses should be reported alongside any operational deployment.
- Truly data-scarce regions may also have noisy meteorological forcings; the framework assumes reasonable input quality.

4.5 Future directions

1. Couple with operational meteorological forecasts (e.g. ERA5, GFS) to extend lead times beyond seven days.
2. Incorporate teleconnection indices (ENSO, PDO) for seasonal-scale predictability.
3. Extend to compound hazards (e.g. heatwave→flash flood) via multi-task outputs.
4. Add physics-informed constraints to the LSTM, as suggested by Kratzert et al. (2019), to ensure mass-balance and energy-balance consistency.
5. Quantify full predictive distributions via quantile regression or Monte-Carlo dropout, replacing the Gaussian-residual mapping used in this preprint.

5. Conclusion

This study integrates regional pre-training, conservative transfer learning, rolling-origin walk-forward validation, Regional Frequency Analysis, and SHAP explainability into a single reproducible framework for hydrological early warning in data-scarce basins. The open-source pipeline that accompanies this manuscript runs end-to-end on synthetic data in minutes (smoke test) and is ready for full-scale CAMELS-US training on a GPU. The conservative fine-tuning recipe — freezing the LSTM cell and training only the dense head — provides a robust and statistically sound approach for adapting regional models to new basins with minimal data. By tying every warning to physical drivers via SHAP and to historical climatology via RFA, the framework supports trustworthy, operationally realistic early warnings that can be deployed in the vast number of watersheds worldwide that lack sufficient observational records.

6. Code and data availability

All code is released under the MIT license at github.com/rajanishankar/hydro_tl_ews. The implementation depends on the open-source NeuralHydrology ([neuralhydrology](https://github.com/rajanishankar/neuralhydrology)) and SHAP ([shap](https://github.com/slundberg/shap)) libraries. CAMELS-US is publicly available from the UCAR/NCAR repository (ral.ucar.edu); USGS NWIS streamflow is accessible via dataretrieval; Daymet meteorological data is available from daymet.ornl.gov.

7. References

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Appendix A · Repository structure

```
configs/    YAML configurations for each pipeline stage
src/hydro_tl_ews/data/  CAMELS / NWIS / synthetic loaders
src/hydro_tl_ews/models/ EA-LSTM cell + differentiable NSE loss
src/hydro_tl_ews/training/ Trainer, transfer recipes, walk-forward backtester
src/hydro_tl_ews/evaluation/ Metrics + Regional Frequency Analysis
src/hydro_tl_ews/xai/   SHAP wrappers
scripts/run_experiment.py CLI entry point
tests/      pytest unit + smoke tests
```